

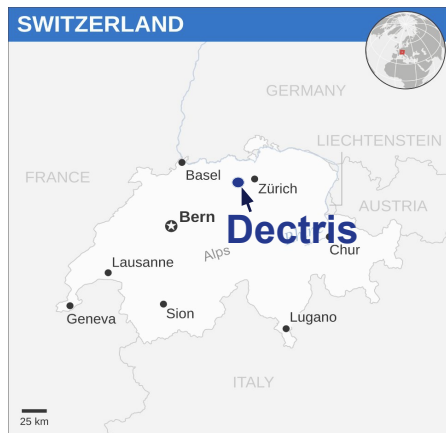
Open-Source EDA as a Strategic Alternative for European Chip Design

Export-control risk at the design-tool layer

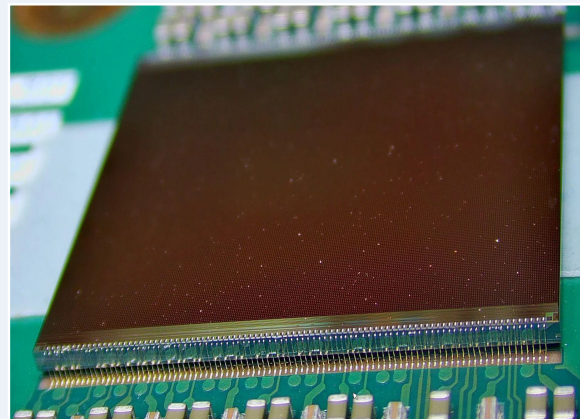
Barbara Wojsz & Tomasz Hemperek

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Who we are



New photon-counting ASIC



Icarus + Yosys + OpenROAD
+
Proprietary tools

Why engineers should care

Export control can interrupt a product even when the design is technically complete.

\$140M

**BIS and DOJ aggregated
penalty against Cadence in
2025**

The case concerned unauthorized exports of EDA hardware, software and design technology. It was not an FDPR ruling.

1 Tool or licence access changes The team may be unable to rerun implementation or signoff.

2 Shipment or customer is blocked Milestones and contracts move immediately.

3 Redesign and requalification ASIC recovery can take months or years.

**EDA is already inside export-control enforcement.
For engineers, the consequence is loss of
continuity.**

What changed: access is no longer only a purchasing

For critical technologies, tool access, export controls and sanctions can change the supply chain.

Old engineering question

Can we buy the tool?



Question companies now need to ask

Are we allowed to use it for this customer and destination?

Can we tape out at this fab?



Does the full route add export-control exposure?

Is the dependency acceptable?



What would happen to the product if access changes?

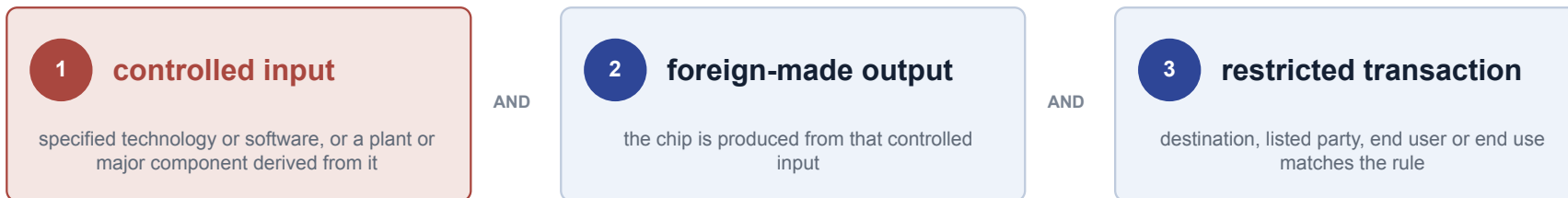
For chip design, the risk can appear before fabrication, at the design-tool layer.

Why the tool can matter

A chip made outside the United States can still fall under US export rules.

FDPR = Foreign Direct Product Rules

Dependency trace from a controlled input to a foreign-made output and a restricted transaction.



Conceptual model

If all three conditions line up, the chip may need a licence or shipment may be blocked.

Why EDA matters

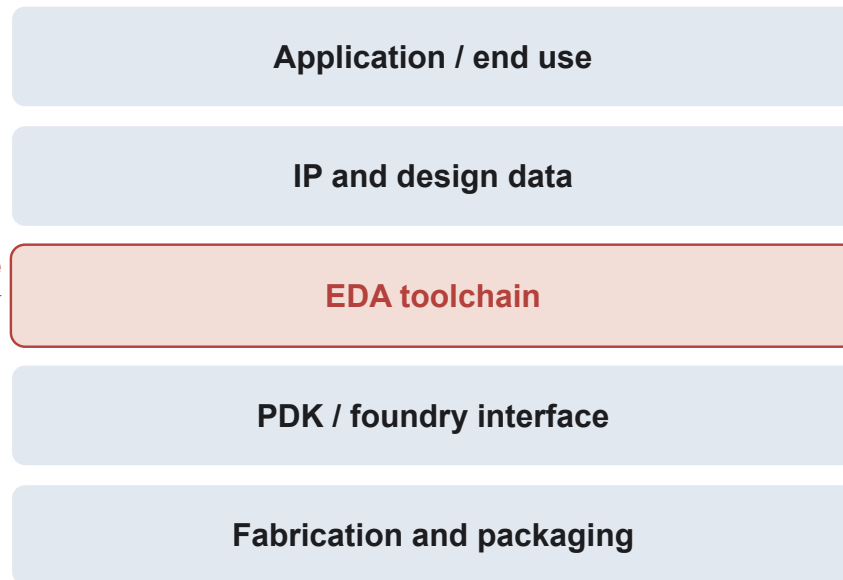
An EDA tool can be part of the controlled input in step 1. The customer, destination and use case decide whether step 3 is triggered.

Important limit: using a US EDA tool is not enough by itself. FDPR is a conditional rule, not an automatic label.

The EDA toolchain is part of the dependency graph

Before fabrication, a chip is simulated, verified, synthesized, placed, routed, and signed off. Using those tools can create additional legal dependencies to foreign export-control law.

risk can enter here



Open-source EDA can change one node in this graph. It does not remove the others.

"Published" Software Exemptions

Software that qualifies as "published" or "publicly available" may fall outside the scope of certain export-control regulations:

EAR (15 C.F.R. § 734.7): Under the US Export Administration Regulations, software or technology that has been made publicly available without restrictions on its further dissemination is considered "published" and is generally not subject to the EAR.

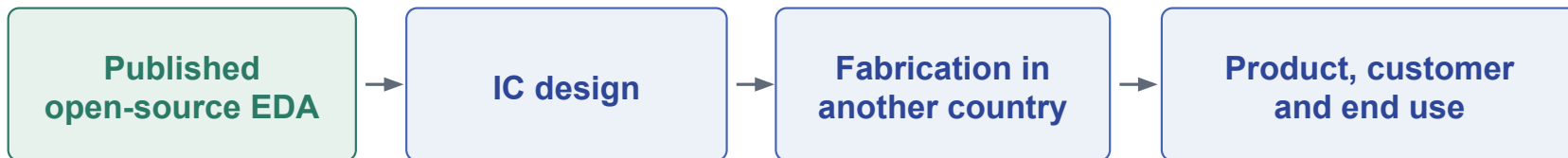
***Exceptions:** Certain categories remain subject to the EAR, including published encryption software classified under ECCN 5D002 (§ 734.7(b)) and 3D printing files for firearms (§ 734.7(c)).*

EU Dual-Use Regulation (Regulation (EU) 2021/821, Annex I – General Software Note): The EU Dual-Use Control List does not control software that is "generally available to the public" or "in the public domain".

***Exceptions:** The General Software Note applies only to Categories 1–9 and does not apply to Category 5, Part 2 ("Information Security").*

One link removed

If the EDA tool is published and outside the applicable controls, that tool does not add export-control exposure to the chip.



The design-tool link is reduced because the published software is outside the relevant control scope.

Remaining dependencies still need analysis

- PDK and standard cells
- Third-party IP
- Cloud and compute services
- Foundry equipment and process technology
- Destination, end user and end use

Where this can work first

Start where open flows are technically credible and the validation problem is bounded.

Mature-node digital blocks

180 nm, 130 nm and 110 nm flows are a realistic entry point because designs are smaller, rules are more stable and commercial fallback is available.

Industrial evidence

Examples already exist: a reported ChipFlow 130 nm BCD test chip and a DECTRIS 110 nm serial-link block.

Support infrastructure

Simulation, CI, waveform tools, cocotb and mixed-signal support can be adopted before replacing signoff.

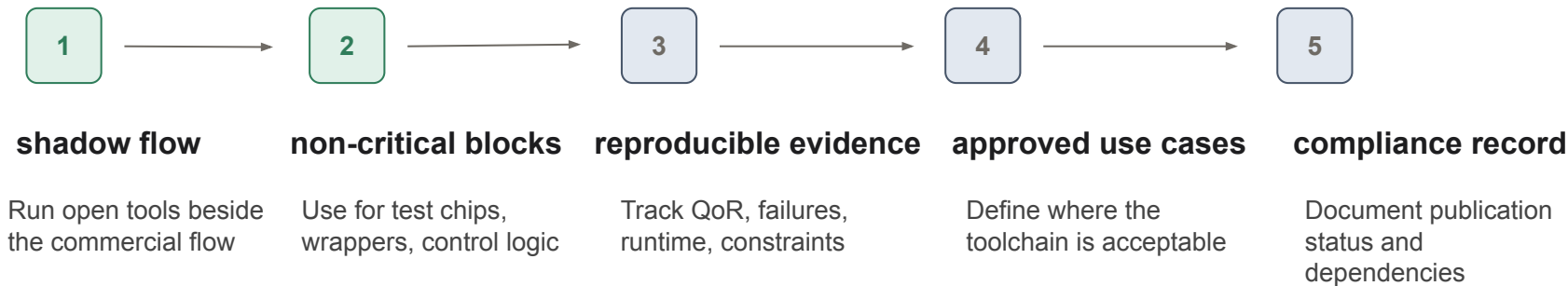
Hard part: PDK translation

The bottleneck is often not RTL. It is LEF, Liberty, layers, DRC assumptions, macros, corners, parasitics and signoff correlation.

Best first target: bounded digital blocks where failures are visible and recovery is cheap.

An adoption path for companies

The objective is not ideological purity. It is to reduce dependence on a single toolchain and keep a alternative available.



New funding rationale — Strategic autonomy gives European funders a reason to pay for maintenance and validation, not prototypes. That money lands upstream: OpenROAD, Yosys, Verilator, open PDKs.

New adopter class — Industrial mature-node users adopt for compliance, not just cost — and compliance-driven users contribute back reproducible flows and signoff evidence.

What the community can do now

Treat open EDA as continuity infrastructure, not only as a cheaper tool.

For maintainers and researchers

Publish complete flows

Include scripts, constraints and tool versions.

Track dependencies

Record PDK, IP, cloud and external services.

Report failures openly

Benchmarks need negative results and reproducibility.

For companies and funders

Fund maintenance

Validation and support matter more than one-off prototypes.

Define approved use cases

Start with bounded industrial workloads.

Connect engineering to policy

Make risk reduction visible in procurement and strategy.

Open EDA cannot remove every export-control risk. It can remove one dependency we can control.

Thank you

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Tomasz Hemperek [linkedin.com/in/hemperek](https://www.linkedin.com/in/hemperek)



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